

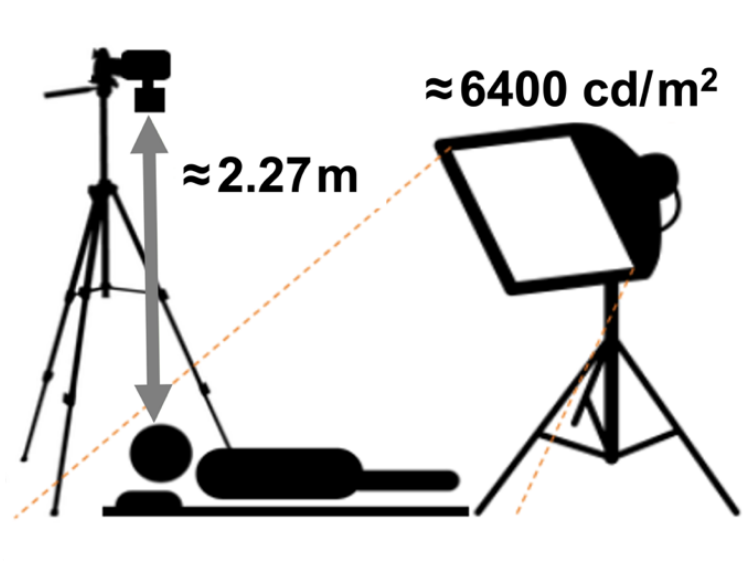
UMINHO-HSFD
Set-up description

Camera orientation Vertical (directed to the floor)

Face orientation Horizontal (directed to the ceiling)

Distance camera-object 2.27 m

Diagram



SET-UP

CUSTOM-BUILT HYPERSPECTRAL IMAGING SYSTEM

Camera	Model	Peltier-cooled low-noise digital monochromatic (Hamamatsu, Model C4742-95-12ER, Hamamatsu Photonics K.K., Hamamatsu City, Japan)
	Resolution	1344 (H) x 1024 (V) pixels
	Precision output	12-bit
	CCD dimension	8.66 x 6.66 mm
	Pixel size	6.45 x 6.45 μ m

Focusing lens	Aperture	f/11
	Zoom	Maximum
	Focal distance	12.5 - 75 mm
	Rendered area per pixel	0.02 ° (1')

CUSTOM-BUILT HYPERSPECTRAL IMAGING SYSTEM	Filters	Fast tunable liquid-crystal filter (placed in front of the camera) (Varispec, Model VS-VIS2-10-HC-35-SQ, Cambridge Research & Instrumentation, Inc., Woburn, Massachusetts, USA)
		Infrared blocking filter
	Wavelength	Range 400 - 720 nm Interval 10 nm Number 33 7 nm at 400 nm Half-height bandwidth 10 nm at 550 nm 16 nm at 720 nm
	Accuracy	Average spectral difference 2 % Average colorimetric error 1.3 units in CIE ΔE_{ab}^* 2000 2.2 unit in CIELAB
ILLUMINATION	Light source	Metal halide lamp (OSRAM, POWERSTAR, HQI-T, 1000W/D, Germany)
	Power	1000 W
	Accessories	UV blocker filter (in front of the lamp)
	Illuminance intensity	6400 cd/m ²
FACES	Position	Fixed
	Selection criteria	Healthy skin
	Exclusion criteria	Skin diseases or anomalies Extensive facial hair
	During acquisition	Cleaned skin (no cream or foundation) Neutral expression with eyes and lips closed Movements reduced with head supporting pillows